

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q11-0050

Kathleen 69 kV
December 31, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0050. The Interconnection Customer proposes a 122.3 MW solar photovoltaic facility interconnecting at the Kathleen 69 kV substation in the North Carolina area. The request was received on June 21, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 44 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$243,054,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Jajome 3403 138-69 kV Autotransformer Bank 2, the Jajome 3403 138 kV/69 kV Auto loads to 131.2% of its 145 MVA rating. Project DFAX: 61.6%.

The Trowbridge 138-69 kV XFMR #1 is overloaded to 129.8% of its 92 MVA emergency rating for the loss of the Bel Air – Home 138 kV. The project contributes 25.3% DFAX (30.9 MW) to this overload.

For the loss of Home – Bel Air Substation 138 kV line, the Trowbridge 138-69 kV Auto loads to 127.8% of its 94 MVA rating. Project DFAX: 25.3%.

The Harbor Beach – Hickory Substation 138 kV is overloaded to 126.0% of its 238 MVA emergency rating for the loss of the Yarnen – Mira Vista 345 kV line. The project contributes 38.7% DFAX (47.3 MW) to this overload.

For the loss of Leavitt Substation – Birchwood Substation 345 kV, the Bear Island Substation – Dunn Siding 345 kV line loads to 120.2% of its 1,303 MVA rating. Project DFAX: 24.5%.

Jajome 3403 138-69 kV Autotransformer Bank 2: 116.1% loading (164 MVA) following the outage of the Jajome 3403 138/69 kV TX T1; DFAX = 61.6%.

Yadkin – No Mans Land 138 kV: 115.9% loading (468 MVA) following the outage of the Mickens – Mira Visa 345 kV; DFAX = 26.5%.

For the loss of Mira Vista – Yarnen 345 kV, the Elmo – Hickory 138 kV loads to 115.9% of its 323 MVA rating. Project DFAX: 38.7%.

For the loss of Mitchell Dam – Skaggs Island Jct 2 Substation 345 kV, the Fritch – Big Hanaford 138 kV loads to 115.8% of its 281 MVA rating. Project DFAX: 8.7%.

The Wheelabrator Portsmouth – Yadkin 138 kV is overloaded to 111.5% of its 334 MVA emergency rating for the loss of the Mickens – Mira Vista Substation 345 kV. The project contributes 12.3% DFAX (15.1 MW) to this overload.

For the loss of Home – Bel Air 138 kV, the No Maries – Trowbridge 138 kV L loads to 111.2% of its 235 MVA rating. Project DFAX: 50.5%.

Phelon Parts 138 kV/69 kV XFMR: 111.1% loading (60 MVA) following the outage of the McCabe – Laurel City Substation 138 kV; DFAX = 49.7%.

The North Towertap – Unknown Substation 138 kV is overloaded to 109.7% of its 277 MVA emergency rating for the loss of the N Tower Tap – Haggin 345 kV. The project contributes 8.6% DFAX (10.5 MW) to this overload.

Home – Bel Air 138 kV: 107.6% loading (258 MVA) following the outage of the Home – No Maries 138 kV Line; DFAX = 52.1%.

For the loss of Haggin – North Tower Tap 345 kV, the Walthall – Unknown 138 kV Line loads to 106.5% of its 272 MVA rating. Project DFAX: 8.6%.

Birchwood – Dist 1000 138 kV: 105.6% loading (304 MVA) following the outage of the Bear Island – Dunn Siding 345 kV; DFAX = 5.6%.

Home Substation – No Maries 138 kV: 105.2% loading (300 MVA) following the outage of the Home – Bel Air 138 kV Line; DFAX = 50.8%.

The Lakeside – Natchez 138 kV line is overloaded to 105.1% of its 197 MVA emergency rating for the loss of the Mitchell Dam – Skaggs Island Jct 2 345 kV L. The project contributes 6.3% DFAX (7.7 MW) to this overload.

Yadkin – Harborgen Substation 138 kV Line: 103.4% loading (222 MVA) following the outage of the Yadkin – Wheelabrator Portsmouth 138 kV Line; DFAX = 15.7%.

The Bel Air – Fabrica Tenere 138 kV is overloaded to 103.3% of its 249 MVA emergency rating for the loss of the Home – No Maries 138 kV L. The project contributes 52.1% DFAX (63.7 MW) to this overload.

Harbor Beach – Ne-Hobbs 138 kV: 103.2% loading (248 MVA) following the outage of the Mira Vista – Yarnen 345 kV; DFAX = 38.7%.

North Tower Tap Substation 345/138 kV Auto: 102.2% loading (438 MVA) following the outage of the N Tower Tap – Haggin 345 kV; DFAX = 13.0%.

For the loss of Bear Island – Dunn Siding 345 kV, the Trenton Diesel – Dist 1000 138 kV loads to 102.1% of its 325 MVA rating. Project DFAX: 5.6%.

Spring St – Chipola 138 kV: 101.7% loading (234 MVA) following the outage of the Mitchell Dam – Haggin Substation 345 kV; DFAX = 6.5%.

For the loss of Mitchell Dam – Skaggs Island Jct 2 345 kV, the Redbed Plains Wind Farm – Westcoast Cellufibre 138 kV loads to 101.4% of its 230 MVA rating. Project DFAX: 6.0%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Jajome 3403 138/69kV TX Bank 1	Replace transformer with higher-rated unit	\$3,672,000	100% (\$3,672,000)
Trowbridge 138-69 kV XFMR #1	Replace transformer with higher-rated unit	\$4,891,000	100% (\$4,891,000)
Trowbridge Substation 138/69kV Auto	Replace transformer with higher-rated unit	\$5,380,000	100% (\$5,380,000)
Hickory – Harbor Beach 138 kV	Rebuild line with higher-capacity conductor	\$10,376,000	100% (\$10,376,000)
Bear Island Substation – Dunn Siding 345 kV line	Rebuild line with higher-capacity conductor	\$97,364,000	100% (\$97,364,000)
Jajome 3403 138/69 kV TX Bank 2	Replace transformer with higher-rated unit	\$6,398,000	100% (\$6,398,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
No Mans Land – Yadkin 138 kV Line	Rebuild line with higher-capacity conductor	\$2,454,000	100% (\$2,454,000)
Elmo – Hickory 138 kV	Rebuild line with higher-capacity conductor	\$2,121,000	100% (\$2,121,000)
Fritch – Big Hanaford 138 kV	Rebuild line with higher-capacity conductor	\$4,049,000	100% (\$4,049,000)
Wheelabrator Portsmouth – Yadkin 138 kV	Rebuild line with higher-capacity conductor	\$4,972,000	100% (\$4,972,000)
No Maries – Trowbridge 138 kV L	Rebuild line with higher-capacity conductor	\$12,730,000	100% (\$12,730,000)
Phelon Parts Substation 138-69 kV Transformer	Replace transformer with higher-rated unit	\$4,718,000	100% (\$4,718,000)
North Tower Tap – Unknown 138 kV L	Reconductor line	\$1,993,000	100% (\$1,993,000)
Bel Air Substation – Home 138 kV	Reconductor line	\$11,031,000	100% (\$11,031,000)
Walthall – Unknown 138 kV	Reconductor line	\$4,726,000	100% (\$4,726,000)
Birchwood – Dist 1000 138 kV	Reconductor line	\$9,679,000	100% (\$9,679,000)
Home Substation – No Maries 138 kV	Reconductor line	\$5,122,000	100% (\$5,122,000)
Lakeside – Natchez 138 kV line	Reconductor line	\$1,330,000	100% (\$1,330,000)
Yadkin – Harborgen Substation 138 kV Line	Reconductor line	\$1,581,000	100% (\$1,581,000)
Bel Air – Fabrica Tenere 138 kV	Reconductor line	\$10,020,000	100% (\$10,020,000)
Harbor Beach – Ne-Hobbs 138 kV L	Reconductor line	\$1,905,000	100% (\$1,905,000)
N Tower Tap 345/138 kV Transformer	Replace transformer with higher-rated unit	\$17,312,000	100% (\$17,312,000)
Trenton Diesel – Dist 1000 138 kV line	Reconductor line	\$15,031,000	100% (\$15,031,000)
Spring St – Chipola 138 kV	Reconductor line	\$2,498,000	100% (\$2,498,000)
Westcoast Cellufibre – Redbed Plains Wind Farm Substation 138 kV	Reconductor line	\$1,701,000	100% (\$1,701,000)
	Total allocated to project		\$243,054,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.